Spectroscopy as a Probe of Spacetime Foam

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Quantum-gravitational spacetime foam imprints an achromatic, stochastic arrival-time jitter on light from distant sources. We show that this jitter modifies the first-order temporal coherence $g^{(1)}(\tau)$ and hence the observed spectrum, providing a spectroscopic probe complementary to existing imaging-channel (PSF, Strehl) constraints. In the dynamical-foam limit the spectrum splits into a coherent line core suppressed by $e^{-\sigma_\phi^2}$ and an incoherent pedestal, yielding a spectral shape-distortion bound on the Ng–Perlman foam exponent α. Applied to the MeV emission lines of GRB 221009A in the dynamical-foam limit ($\tau_{\rm foam} \ll \tau_c$) we obtain $\alpha \gtrsim 0.63$, a weak, lever-arm-dominated floor. For the broadband CMB the foam induces an $e^{-\sigma_\phi^2(\nu)}$ deficit ($\sigma_\phi^2 \propto \nu^2$) with a low-frequency pedestal; a fit to the COBE/FIRAS monopole spectrum reaches $\alpha \gtrsim 0.52$ (95% CL) under the physical Planck-scale pedestal kernel, and next-generation spectrometers (PIXIE, SuperPIXIE, Voyage 2050) are forecast to reach $\alpha \gtrsim 0.54$–0.55, the bound saturating logarithmically as the foam deficit is nearly degenerate with a temperature shift. The pedestal-independent line-width bound does not exclude the holographic value $\alpha = 2/3$; the TeV bound, if accepted at face value (conditional on the pedestal kernel shape), would exclude it. Resolving whether the holographic model survives requires a confirmed narrow spectral line at $E_{\rm obs} \gtrsim 720\,{\rm MeV}$ in a $z \sim 0.15$ GRB, which would promote the line-width bound into the exclusion regime independently of the pedestal assumption.

I. INTRODUCTION

A long-standing expectation from the union of quantum mechanics and general relativity is that the smooth pseudo-Riemannian manifold of classical spacetime breaks down at the Planck length $\ell_{\rm P} = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m. In Wheeler's original "spacetime foam" picture [1], quantum fluctuations of the metric on this scale render the geometry stochastic; similar conclusions are reached in the Euclidean path-integral treatment [2] and in modern holographic and entropic-gravity arguments from very different starting points. While a complete theory of quantum gravity remains elusive, the question of whether residual imprints of Planck-scale geometric fluctuations can accumulate over cosmological baselines and become observable is one of the few empirical handles on physics in the Planck regime, and it has motivated a substantial body of phenomenological work over the past two decades [3, 4].

A widely studied parametrization due to Ng and collaborators [4–6], the α-foam model, encodes the foam dynamics in a power-law path-length variance

$$\sigma_\ell^2(D) = A_\alpha\, D^{2(1-\alpha)}\, \ell_{\rm P}^{2\alpha}, \tag{1}$$

where D is the line-of-sight comoving distance to the source — the appropriate measure rather than the luminosity distance, since the phase accumulation is a property of spacetime itself rather than the photon path [4] — α parametrizes the foam model, and A_α is a dimensionless coefficient of order unity. A photon traversing distance D passes through approximately $N \sim D/\ell_{\rm P}$ independent Planck-scale spacetime cells, each imparting a tiny random displacement to the photon path; the exponent α controls how the per-cell displacements accumulate: for random-walk foam ($\alpha = 1/2$), contributions add in quadrature ($\sigma_\ell \propto \sqrt{D\ell_{\rm P}}$); holographic foam ($\alpha = 2/3$) grows more slowly, motivated by the heuristic that the information in a spatial region scales with its bounding area rather than its volume — an analogy to the holographic entropy bound; and the weak-foam limit ($\alpha = 1$) yields a distance-independent $\sigma_\ell \propto \ell_{\rm P}$, corresponding to perfectly correlated cell contributions.

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The primary observational probe of stochastic foam to date has been the imaging channel. Light propagating through spacetime foam accumulates a phase dispersion $\Delta\phi \simeq 2\pi\ell^{1-\alpha}\ell_{\rm P}^{\alpha}/\lambda$ across the entrance aperture, degrading PSF sharpness and erasing interference fringes once $\Delta\phi \gtrsim 2\pi$ [7]. For PKS 1413+135 ($\ell = 1.216$ Gpc) observed with HST at $\lambda = 1.6\,\mu$m, the random-walk model ($\alpha = 1/2$) predicts $\Delta\phi \sim 10 \times 2\pi$, so the detection of Airy rings in this image marginally rules out that model; the holographic model ($\alpha = 2/3$) predicts $\Delta\phi \sim 10^{-9} \times 2\pi$ and is obviously not ruled out [4, 8, 9]. Models with long-range transverse correlations evade even the random-walk bound [10].

The most luminous GRB ever recorded, GRB 221009A at $z = 0.151$ [11], has been examined in the imaging channel by Steinbring [12] and Steinbring & Ng [13]. Using a multiwavelength average of foam-induced angular broadening — analogous to a seeing disk from atmospheric turbulence — Steinbring [12] showed that the holographic ($\alpha = 2/3$) model predicts an angular widening below $\sim 1^\circ$, consistent with the Fermi and Swift localization data for GRBs at $z \leq 5$ up to peak energies of $\sim 10\,$MeV. Steinbring & Ng [13] extended this to the full multiwavelength follow-up of GRB 221009A — from optical and near-infrared through X-rays and the highest-energy gamma-rays ever attributed to a GRB — computing the holographic-foam-widened PSF for real telescopes and finding all data consistent with $\alpha = 2/3$. These imaging-channel analyses show that the holographic model is not excluded by the spatial localization of GRB 221009A.

We develop a framework based on the first-order temporal coherence function $g^{(1)}(\tau)$ of light from astrophysical sources. This function is complementary to the first-order spatial coherence function $g^{(1)}(B)$, which forms the basis of imaging-channel probes. The first-order coherence — the field–field autocorrelation — inherits a foam suppression factor that decreases continuously from unity toward complete fringe erasure as the accumulated phase variance σ_ϕ^2 grows. In our framework the phase variance arises from temporal fluctuations of foam, whereas in the imaging channel it arises from spatial fluctuations; correspondingly, foam imprints itself on the spectrum rather than on the spatial brightness profile.

The remainder of the paper is organized as follows. Section II introduces the spacetime foam model: the arrival-time jitter parameterization, the temporal autocorrelation $C_\alpha(\tau)$, and the distribution of differential jitter. Section III derives the first-order coherence function $g^{(1)}_{\rm obs}(\tau)$ of the observed field and establishes the Gaussian closed form. Section IV translates these results into spectroscopic observables via the Wiener–Khinchin theorem, establishing the split of the observed spectrum into a coherent line core suppressed by $e^{-\sigma_\phi^2}$ and a broad pedestal, and connects them to the imaging Strehl ratio. Section V applies the spectral shape-distortion bound to two sources: GRB 221009A (Section V A) and the CMB as observed by COBE/FIRAS (Section V B). Section VI summarizes the main results and outlines future observational directions.

II. SPACETIME FOAM MODEL

A. Arrival-Time Jitter

We adopt the Ng–Perlman picture of spacetime foam: stochastic, achromatic path-length fluctuations whose cumulative effect along a line of sight is an arrival-time jitter $\delta t(t)$, which a carrier reads as a phase jitter σ_ϕ. Following the prior foam literature we take a point source on a single line of sight; at the level of first-order coherence an extended incoherent source is the linear superposition of such points. The transverse behavior of the foam across an extended source is beyond our scope.

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\rm obs}(t) = E_{\rm src}(t - \delta t(t))\,, \tag{2}$$

where $E_{\rm src}(t)$ is a single mode of a stationary, zero-mean complex Gaussian (chaotic) field — appropriate for the thermal and incoherent sources here (GRB high-energy emission, CMB) — and $\delta t(t)$ is the foam jitter, a stationary process independent of $E_{\rm src}$. Here t is the observer's time, with the deterministic light-travel delay and cosmological time dilation absorbed into the argument of $E_{\rm src}$, so $\delta t(t)$ is the zero-mean foam fluctuation about the mean arrival time. The jitter arises from cumulative path-length fluctuations along the line of sight of propagation distance D:

$$\delta t(t) = \frac{1}{c}\,\Delta\ell_{\rm tot}(t). \tag{3}$$

Because the path-length fluctuation is geometric—arising from stochastic metric perturbations of the FLRW background rather than a dispersive medium—δt is achromatic (frequency-independent), distinguishing foam from plasma dispersion, interstellar scintillation, and standard spectral distortions [4]. A monochromatic carrier of observed fre-

quency $\omega_0 = 2\pi\nu_0$ reads the same delay as a phase jitter $\delta\phi = \omega_0\,\delta t$, giving a phase variance

$$\sigma_\phi^2 \;=\; \left(\frac{2\pi}{\lambda_{\rm obs}}\right)^2 \sigma_\ell^2(D_C), \tag{4}$$

on which all spectroscopic bounds below rest.

B. Temporal Decorrelation of Spacetime Foam

The Ng–Perlman variance $\sigma_\ell^2(D)$ (Eq. (1)) fixes the *amplitude* of the path-length fluctuations along a single ray but not their *temporal* correlation. Quantum-foam models posit that each Planck-scale cell fluctuates on a characteristic timescale $\tau_{\rm foam}$, so the jitter at times t and $t+\tau$ is correlated for $\tau \ll \tau_{\rm foam}$ and essentially independent for $\tau \gg \tau_{\rm foam}$. We encode this in the normalized *temporal* autocorrelation $C_\alpha(\tau; D)$ of the integrated path-length fluctuation $\Delta\ell_{\rm tot}(t)$,

$$\big\langle \Delta\ell_{\rm tot}(t)\,\Delta\ell_{\rm tot}(t+\tau)\big\rangle = \sigma_\ell^2(D)\,C_\alpha(\tau; D), \tag{5}$$

normalized so that $C_\alpha(0; D) = 1$. Because C_α is defined on the *integrated* path, its temporal shape inherits a residual D-dependence from the way the per-cell foam dynamics over the $\sim D/\ell_{\rm P}$ cells crossed by the line of sight combine; the explicit D-argument records this dependence.

Since no physical source is pointlike, the underlying foam correlation is spatio-temporal, $C(\mathbf{r}, \tau)$, and we read $C_\alpha(\tau; D)$ as the *effective* temporal correlation left after coarse-graining over the source's emitting four-volume and propagation path. The reduction to a purely temporal C_α assumes the spatio-temporal correlation factorizes, $C(\mathbf{r}, \tau) \approx C_{\rm sp}(\mathbf{r})\,C_\alpha(\tau)$, with the spatial factor absorbed into the normalization; we flag this as an assumption, since the spatial correlation structure over cosmological path lengths is not constrained here.

The timescale $\tau_{\rm foam}$ is an additional free parameter of the foam model, not fixed by the α-model variance relation alone. In the standard phenomenological framework [6, 14], the foam refreshes on the Planck time $\tau_P \equiv \ell_P/c \approx 5 \times 10^{-44}$ s, so the integrated path-length jitter $\Delta\ell_{\rm tot}(t)$ — a sum of $\sim D/\ell_P$ independent contributions — likewise decorrelates on timescale $\tau_{\rm foam} \sim \tau_P$. This is the *dynamical-foam* regime adopted throughout: $\tau_{\rm foam}$ is far shorter than any timescale on which any observable coherence has support, so $C_\alpha(\tau) \to 0$ there. By contrast, models with long-range *temporal* correlations — the temporal analogue of the long-range *transverse* spatial correlations of Ref. [10] that evade interferometric imaging bounds — would give $C_\alpha(\tau) \approx 1$ over macroscopic lags and drive $\sigma_\Delta^2(\tau) \to 0$, suppressing the foam signal; the bounds derived below therefore rest on the dynamical-foam assumption.

C. Differential Jitter and Its Distribution

We now define the differential jitter, which enters the coherence integrals, between photons arriving at times t and $t+\tau$:

$$\Delta(\tau) \equiv \delta t(t+\tau) - \delta t(t). \tag{6}$$

Expanding the square and applying Eq. (5):

$$\begin{aligned}
\sigma_\Delta^2(\tau) &= \frac{1}{c^2}\big\langle [\Delta\ell_{\rm tot}(t+\tau) - \Delta\ell_{\rm tot}(t)]^2 \big\rangle \\
&= \frac{1}{c^2}\left[\sigma_\ell^2 + \sigma_\ell^2 - 2\sigma_\ell^2\,C_\alpha(\tau)\right] \\
&= \frac{2\sigma_\ell^2(D)}{c^2}\left[1 - C_\alpha(\tau; D)\right].
\end{aligned} \tag{7}$$

By the central limit theorem (CLT), $\Delta(\tau)$ is a sum of contributions from many foam cells and is therefore Gaussian with zero mean:

$$\Delta(\tau) \sim \mathcal{N}\big(0,\, \sigma_\Delta^2(\tau)\big)\,, \tag{8}$$

where σ_Δ^2 is taken from Eq. (7); for any source at $D \gtrsim 1$ kpc, the cell count $N_{\rm cells} \sim D/\ell_{\rm P} \gtrsim 10^{54}$ amply satisfies the large-N requirement and higher-cumulant corrections are negligible. For $\alpha = 1/2$ (random walk), per-cell path-length variances are equal ($\ell_{\rm P}^2$ each) and the CLT applies in its standard form. For $\alpha \neq 1/2$, the per-cell contributions scale differently with D and the equality of per-cell variances is a model assumption rather than a direct CLT consequence; Gaussianity is adopted as the minimal assumption consistent with the α-model.

III. FIRST-ORDER COHERENCE

A. Jitter Convolution Formula for $g^{(1)}$

The observed first-order coherence is

$$g^{(1)}_{\rm obs}(\tau) = \frac{\langle E^*_{\rm obs}(t)\, E_{\rm obs}(t+\tau)\rangle}{\langle |E_{\rm obs}(t)|^2\rangle}. \tag{9}$$

Substituting Eq. (2) and letting $u = t - \delta t(t)$, $v = t + \tau - \delta t(t+\tau)$, so that $v - u = \tau - \Delta(\tau)$, stationarity of $E_{\rm src}$ gives

$$\langle E^*_{\rm src}(u)\, E_{\rm src}(v)\rangle = \langle |E_{\rm src}|^2\rangle\, g^{(1)}_{\rm src}(\tau - \Delta(\tau)). \tag{10}$$

Since δt is independent of $E_{\rm src}$ and $\langle |E_{\rm obs}|^2\rangle = \langle |E_{\rm src}|^2\rangle$ by stationarity (so the Eq. (9) normalization cancels), averaging over the jitter yields

$$g^{(1)}_{\rm obs}(\tau) = \big\langle g^{(1)}_{\rm src}(\tau - \Delta(\tau))\big\rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma^2_\Delta(\tau))\, g^{(1)}_{\rm src}(\tau - \Delta)\, d\Delta. \tag{11}$$

For a quasi-monochromatic source $g^{(1)}_{\rm src}(\tau) = e^{-i\omega_0\tau}\,\gamma(\tau)$ with γ the slowly varying envelope coherence, Eq. (11) reads

$$g^{(1)}_{\rm obs}(\tau) = e^{-i\omega_0\tau}\,\big\langle e^{\,i\omega_0\Delta(\tau)}\,\gamma(\tau - \Delta(\tau))\big\rangle_{\delta t}, \tag{12}$$

the envelope responding to the achromatic differential delay $\Delta(\tau)$ (variance σ^2_Δ) and the carrier acquiring the phase $\omega_0\Delta(\tau)$. The carrier suppression is the characteristic function of that phase,

$$\mathcal{F}_{\rm foam}(\tau) \equiv \big\langle e^{\,i\omega_0\Delta(\tau)}\big\rangle_{\delta t} = e^{-\omega_0^2\sigma^2_\Delta(\tau)/2} = e^{-\sigma^2_\phi\,[1 - C_\alpha(\tau; D)]}, \tag{13}$$

with $\mathcal{F}_{\rm foam}(0) = 1$, the differential phase variance $D_\phi(\tau) \equiv \omega_0^2\sigma^2_\Delta(\tau) = 2\sigma^2_\phi[1 - C_\alpha(\tau; D)]$, and σ^2_ϕ from Eq. (4); at $\tau \to \infty$ ($C_\alpha \to 0$), $\mathcal{F}_{\rm foam} \to e^{-\sigma^2_\phi}$.

B. Gaussian Source Profile

The GRB 221009A emission line is well described by a Gaussian profile in each epoch [15], and it serves as the representative case admitting the simplest closed form. A Gaussian line of 1σ width σ_ν centered on ν_0 corresponds via the Wiener–Khinchin theorem to the coherence function

$$g^{(1)}_{\rm src}(\tau) \;=\; e^{-i\omega_0\tau}\, e^{-\tau^2/2\tau_G^2}, \tag{14}$$

where $\omega_0 \equiv 2\pi\nu_0$ and $\tau_G \equiv 1/(2\pi\sigma_\nu)$ is the Gaussian coherence decay time. Substituting the Gaussian profile (14) into the jitter-convolution integral (12) and evaluating the resulting Gaussian integral over Δ (with $\sigma^2_\Delta \equiv \sigma^2_\Delta(\tau)$) gives the exact closed form

$$\begin{aligned} g^{(1)}_{\rm obs}(\tau) = \frac{\tau_G}{\sqrt{\tau_G^2 + \sigma^2_\Delta(\tau)}}\, \exp\!\left[-i\omega_0\tau + \frac{i\,\omega_0\,\sigma^2_\Delta(\tau)\,\tau}{\tau_G^2 + \sigma^2_\Delta(\tau)}\right] \\ \times \exp\!\left[-\frac{\tau^2}{2\big[\tau_G^2 + \sigma^2_\Delta(\tau)\big]}\right] \exp\!\left[-\frac{D_\phi(\tau)}{2} + \frac{\omega_0^2\sigma^4_\Delta(\tau)}{2\big[\tau_G^2 + \sigma^2_\Delta(\tau)\big]}\right]. \end{aligned} \tag{15}$$

The observed coherence is again a Gaussian-modulated oscillation, but with three foam-induced modifications: (i) a slight shift of the carrier, $\nu_0^{\rm eff} = \nu_0\big[1 - \sigma^2_\Delta(\tau)/(\tau_G^2 + \sigma^2_\Delta(\tau))\big]$; (ii) a wider Gaussian decay of $g^{(1)}_{\rm obs}(\tau)$ in lag (decay scale $\sqrt{\tau_G^2 + \sigma^2_\Delta(\tau)}$ vs. τ_G); and (iii) a global amplitude suppression set by the differential phase variance $D_\phi(\tau) = \omega_0^2\sigma^2_\Delta(\tau)$ (Eq. (13)), at the carrier ω_0. The marginal phase variance $D_\phi = \omega_0^2\sigma^2_\Delta$ is exact, while the carrier shift (i) and the $\omega_0^2\sigma^4_\Delta$ amplitude term are carrier–envelope coupling of order σ^2_Δ/τ_G^2. In the slowly-varying-envelope limit $\sigma_\Delta \ll \tau_G$ these vanish and $\mathcal{F}_{\rm foam} = e^{-D_\phi/2}$ factors out cleanly, $g^{(1)}_{\rm obs} \to \mathcal{F}_{\rm foam}(\tau)\, g^{(1)}_{\rm src}(\tau)$ — the pure multiplicative suppression on which the spectroscopic bounds rest. Normalization $g^{(1)}_{\rm obs}(0) = 1$ holds because $\sigma_\Delta(0) = 0$ (hence $D_\phi(0) = 0$). The narrowband limit $\tau_G \to \infty$ of Eq. (15) reduces to a pure phase-modulated oscillation, which is the single-mode building block from which the broadband CMB spectrum of Section V B 2 is constructed by incoherent superposition.

IV. IMPLICATIONS FOR SPECTROSCOPY

A. The Wiener–Khinchin Bridge

The foam-modified observed coherence function — given in general by the jitter-convolution integral Eq. (11) with the closed form Eq. (15) — carries direct spectroscopic content: for a stationary field the power spectral density $S(\nu)$ and $g^{(1)}(\tau)$ are Fourier-transform pairs, implying

$$S_{\rm obs}(\nu) = \mathcal{F}\Big\{ g^{(1)}_{\rm obs}(\tau) \Big\}\,(\nu). \tag{16}$$

The theorem applies because astronomical emission lines have finite linewidth (so the Fourier transform is well defined) and because the foam-induced jitter is a stationary process independent of the source (foam acts on light after emission, while $\delta t(t)$ statistics are time-translation invariant on observational timescales). The observed field is then stationary when the integration time exceeds the foam coherence time $\tau_{\rm foam}$ (the lag scale beyond which $C_\alpha(\tau; D) \to 0$), so that the time-averaged spectrum coincides with its ensemble average: $T_{\rm int} \gg \tau_{\rm foam}$.

In the dynamical limit $C_\alpha(\tau; D) \to 0$ for $|\tau| \gtrsim \tau_{\rm foam}$, $\mathcal{F}_{\rm foam}(\tau)$ acquires a long-lag plateau $e^{-\sigma_\phi^2}$ outside a narrow window $|\tau| \lesssim \tau_{\rm foam}$. Fourier-transforming via Eq. (16), the plateau produces a coherent core $e^{-\sigma_\phi^2} S_{\rm src}(\nu)$, while the short-lag window redistributes the complementary fraction $1 - e^{-\sigma_\phi^2}$ of the source flux into a broad pedestal. This core-plus-pedestal split holds for any source, with σ_ϕ^2 evaluated frequency by frequency ($\sigma_\phi^2 \to \sigma_\phi^2(\nu)$) for a broadband source; its broadband form for the CMB, including the pedestal shape, is worked out for a specific model of the temporal autocorrelation C_α in Section V B 2.

For the Gaussian source profile of Section III B, Fourier-transforming Eq. (15) with σ_Δ^2 held at its asymptotic value $2\sigma_\ell^2/c^2$ gives the explicit Gaussian core

$$S^{\rm core}_{\rm obs}(\nu) \;=\; \frac{\tau_G}{\sqrt{\tau_G^2 + \sigma_\Delta^2}}\, \exp\!\left[-\frac{\sigma_\phi^2\, \tau_G^2}{\tau_G^2 + \sigma_\Delta^2} \right] \cdot \frac{1}{\sqrt{2\pi}\, \sigma_{\rm obs}}\, \exp\!\left[-\frac{(\nu - \nu_0^{\rm eff})^2}{2\sigma_{\rm obs}^2} \right], \tag{17}$$

with $\nu_0^{\rm eff} = \nu_0 \tau_G^2/(\tau_G^2 + \sigma_\Delta^2)$ and $\sigma_{\rm obs} = 1/[2\pi\sqrt{\tau_G^2 + \sigma_\Delta^2}] < \sigma_\nu$; the result is exact for the single-carrier Gaussian source of Eq. (15). The Gaussian's carrier shift, line narrowing, and peak suppression are the frequency-domain images of items (i)–(iii) of Section III B.

The complementary $1 - e^{-\sigma_\phi^2}$ of the line flux reappears as a pedestal: in the regime $\sigma_\Delta \ll \tau_G$, the plateau/residual decomposition of $\mathcal{F}_{\rm foam}$ sends the short-lag residual to the pedestal

$$\begin{aligned} S^{\rm ped}_{\rm obs}(\nu) &= \int_{-\infty}^{\infty} d\tau\, e^{\,i2\pi(\nu-\nu_0)\tau}\, e^{-\tau^2/2\tau_G^2} \left[e^{-\sigma_\phi^2[1 - C_\alpha(\tau)]} - e^{-\sigma_\phi^2} \right] \\ &\xrightarrow{\tau_{\rm foam} \ll \tau_G} \hat{R}(\nu - \nu_0) = \tau_{\rm foam}\, \hat{h}\big[(\nu - \nu_0)\, \tau_{\rm foam}\big], \end{aligned} \tag{18}$$

where the limit rescales $\tau = \tau_{\rm foam} x$, $h(x) \equiv e^{-\sigma_\phi^2[1 - C_\alpha(\tau_{\rm foam} x)]} - e^{-\sigma_\phi^2}$, and $\hat{h}(y) \equiv \int_{-\infty}^{\infty} dx\, e^{\,i2\pi x y}\, h(x)$ is a fixed $\mathcal{O}(1)$ function of unit-order width; the full observed line is $S_{\rm obs}(\nu) = S^{\rm core}_{\rm obs}(\nu) + S^{\rm ped}_{\rm obs}(\nu)$. The pedestal width $\Delta\nu_{\rm ped} \sim 1/\tau_{\rm foam}$ exceeds the line width $\sigma_\nu = 1/(2\pi\tau_G)$ by $\tau_G/\tau_{\rm foam} \gg 1$ (Fourier reciprocity: narrow in lag, broad in frequency), and its total weight $\int S^{\rm ped}_{\rm obs}\, d\nu = 1 - e^{-\sigma_\phi^2}$ complements the core's $e^{-\sigma_\phi^2}$, conserving line photon number. Spread over $\sim 1/\tau_{\rm foam}$ — for Planck-scale foam $\Delta\nu_{\rm ped} \sim 10^{43}\,$Hz — the pedestal's in-line contribution is suppressed by $\tau_{\rm foam}/\tau_G \ll 1$, so the spectrum near the line peak is dominated by the coherent core.

The core and pedestal differ in predictive status. The suppression $e^{-\sigma_\phi^2(\nu)}$ is exact for any σ_ϕ^2 — from Gaussian phase statistics and long-lag decorrelation ($C_\alpha \to 0$) — so the linearization $1 - e^{-\sigma_\phi^2} \simeq \sigma_\phi^2$ is only a convenience. The observed deficit equals $1 - e^{-\sigma_\phi^2(\nu)}$ while the pedestal stays below the core. Being broad ($\sim 1/\tau_{\rm foam}$), the pedestal returns only a fraction $f_{\rm band} \sim \Delta\nu\, \tau_{\rm foam}$ to a band $\Delta\nu$, so the core exceeds this floor for $\sigma_\phi^2 \lesssim -\ln f_{\rm band}$ — deep into the optically thick regime, the more so as self-convolution broadens the pedestal. Beyond it the deficit saturates at the floor: optically thick foam remains evident, but its strength requires the pedestal profile, the C_α-dependent self-convolution series. Photon-number conservation preserves the qualitative signature throughout.

B. Connection to the Imaging Strehl Ratio

The first-order coherence function $g^{(1)}_{\rm obs}(\tau)$ derived above describes *temporal* coherence along a single line of sight, with the foam suppression governed by the temporal decorrelation $C_\alpha(\tau; D)$ of Eq. (13). The Strehl ratio of an imaging

telescope is the *spatial* counterpart: phase variations across the aperture plane are governed by an aperture-plane coherence function $C_\alpha^{\rm sp}(B;D)$, constructed by direct analogy to the temporal $C_\alpha(\tau;D)$ of Section II B under the same foam-cell statistics.

The structural equivalence is not merely formal: in both cases the foam suppression arises as the characteristic function of a zero-mean Gaussian differential random variable. Temporally, the variable is the differential jitter $\Delta(\tau) = \delta t(t+\tau) - \delta t(t)$ with variance $\sigma_\Delta^2(\tau) = (2\sigma_\ell^2/c^2)[1 - C_\alpha(\tau;D)]$ (Eq. (7)). Spatially, the variable is the differential phase $\Delta\phi(B) = \phi(\mathbf{x}+\mathbf{B}) - \phi(\mathbf{x})$ between two points on the aperture plane separated by baseline B, with variance

$$\sigma_{\Delta\phi}^2(B) \;=\; 2\sigma_\phi^2\big[1 - C_\alpha^{\rm sp}(B;D)\big]. \tag{19}$$

The focal-plane field of a telescope is the Fraunhofer Fourier transform of the aperture-plane field, giving the diffraction-limited PSF when the aperture phase is uniform. Foam imprints a Gaussian random phase $\phi(\mathbf{x})$ on the aperture with variance σ_ϕ^2 and two-point correlation $C_\alpha^{\rm sp}(B;D)$, blurring this PSF. In the resolved limit $C_\alpha^{\rm sp}(D_{\rm tel};D) \to 0$ — where aperture phases at distinct points decorrelate — the on-axis peak is attenuated by the Strehl ratio

$$S \;\simeq\; e^{-\sigma_\phi^2} \tag{20}$$

(extended-Maréchal [4]), the surrounding Airy fringes are washed out by the same factor, and the complementary $1 - e^{-\sigma_\phi^2}$ of the source flux scatters into an incoherent halo.

Imaging and spectroscopic channels each probe σ_ϕ^2 through a Fourier transform — aperture position $\leftrightarrow$ focal-plane angle in imaging (Fraunhofer), lag $\leftrightarrow$ frequency in spectroscopy (Wiener–Khinchin) — with the rapid chromatic tightening $\sigma_\phi \propto 1/\lambda$ exploited in both Ng & Perlman's [4] Table 1 and the analyses of Section V A. The shared σ_ϕ^2 that suppresses the coherent diffraction-limited PSF in imaging also suppresses the coherent line core in spectroscopy, placing the published imaging constraints [4, 7–9] on a common footing with the present framework.

Two caveats temper the duality. A quantitative cross-channel comparison is model-dependent: it requires relating the spatial ($C_\alpha^{\rm sp}$) and temporal (C_α) foam correlation functions under a common foam-cell model, so the constraints coincide only to the precision of those shared details. And the channels differ in their reference truth: in imaging the aperture is known a priori, whereas in spectroscopy the intrinsic source spectrum is itself uncertain, making the foam contribution harder to isolate.

V. SOURCES

Time-varying spacetime foam modifies the source first-order coherence, and hence its spectrum, in a wavelength-dependent way through the scaling $\sigma_\phi \propto 1/\lambda$. Two observable consequences follow. First, spectral features distort per Eq. (17) with a carrier shift, line narrowing, and peak suppression scaling as $\sigma_\phi \propto 1/\lambda$. Although the absolute width narrows (item (ii), Section III B), the center shifts to lower frequency by a larger factor (item (i)), so the *fractional* width σ/ν broadens — the quantity constrained below. Second, the broadband spectral energy distribution is distorted: the cosmic microwave background, expected to be a near-perfect blackbody, would acquire foam-induced deviations whose chromatic signature is distinct from standard chemical-potential and Compton-y distortions.

A. GRB 221009A

1. *The source*

GRB 221009A ("the BOAT," Brightest Of All Time) is the most energetic gamma-ray burst in the instrumental record, with isotropic-equivalent energy $E_{\rm iso} \sim 10^{55}\,$erg at redshift $z = 0.151$ [11], a once-in-$\sim 10^4$-year event [16]. Adopting the Planck 2018 flat ΛCDM cosmology used throughout [17] ($\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$, $H_0 = 67.36\,{\rm km\,s^{-1}\,Mpc^{-1}}$), the comoving distance is $D_C(z=0.151) \approx 0.65\,{\rm Gpc} \approx 2.0\times10^{25}\,{\rm m}$ and $\ln(D_C/\ell_P) \approx 138.4$.

Two features of this event make it a unique foam laboratory. First, LHAASO detected more than 5000 gamma rays above 0.5 TeV (WCDA) [18] and 142 events above 3 TeV (KM2A) extending to $\sim$13 TeV [19]; these provide an absolute-flux test of the foam suppression. Second, two independent analyses report a narrow gamma-ray emission feature in the Fermi/GBM data, interpreted as blueshifted electron–positron pair annihilation [11, 15], at observed energies spanning 6–37 MeV. Zhang et al. [15] detect a Gaussian line whose centroid evolves as $E_{\rm line} \propto t^{-1}$ from $\approx 37\,$MeV to $\approx 6\,$MeV, with a fractional width $r_{\rm obs} \equiv \sigma_{\rm obs}/E_{\rm line} = 0.10 \pm 0.01$ constant across seven well-constrained sub-intervals spanning the 9–37 MeV lever arm ($\chi^2/{\rm dof} = 5.3/6$); the foam analysis depends only on the Doppler-invariance of the line's fractional width, not on the emission mechanism.

These two data sets yield two complementary foam probes that draw on the same $\sigma_\phi^2 \propto \nu^2$ physics but differ sharply in strength and in what they assume. The stronger is an *absolute* bound: the mere survival of TeV photons bounds σ_ϕ^2 directly. The weaker is a *relative* one: energy-dependent broadening of the MeV line constrains α only in the deep strong-foam regime. Both operate in the dynamical-foam regime $\tau_{\rm foam} \ll \tau_c$ adopted throughout Section II; the opposite quasi-static limit is addressed explicitly as Assumption (i) below.

2. *Survival of the LHAASO TeV photons*

The foam-modified spectrum of a broadband source splits into a coherent core and an incoherent pedestal,

$$S_{\rm obs}(\nu) = e^{-\sigma_\phi^2(\nu)}\, S_{\rm src}(\nu) + \int S_{\rm src}(\nu')\, \hat{R}(\nu-\nu';\nu')\, d\nu', \tag{21}$$

where the first term is the surviving carrier, suppressed per mode by $e^{-\sigma_\phi^2(\nu)}$, while the second is the flux removed from the core, $1-e^{-\sigma_\phi^2}$, redistributed by the pedestal kernel $\hat{R}$ of width $\sim 1/\tau_{\rm foam}$. Two assumptions govern whether the pedestal returns flux to the observing band.

Assumption (i): dynamical foam. The pedestal exists only if foam decorrelates, $C_\alpha(\tau) \to 0$; the quasi-static limit $C_\alpha \to 1$ yields $\mathcal{F}_{\rm foam} \to 1$ with no suppression to constrain (Section II B). For Planck-time-refresh foam ($\tau_{\rm foam} \sim \tau_P \equiv \ell_P/c \approx 5\times10^{-44}\,$s), this escape closes at TeV: with $\tau_c/\tau_{\rm foam} \gtrsim 10^{22}$ *even at* $37\,MeV$ and $\tau_c \propto 1/\nu$, the margin grows toward TeV. This energy-scaling argument does not address models with long-range temporal correlations (the temporal analogue of Ref. [10]; see Section II B), which give $C_\alpha \approx 1$ regardless of photon energy and remain a separate escape from all bounds in this paper.

Assumption (ii): bandwidth $\ll$ pedestal. The pedestal width is $\sim c/\ell_P \approx 10^{43}\,$Hz, whereas the entire electromagnetic output of a GRB (radio to TeV) spans $\lesssim 10^{27}\,$Hz, a ratio $\sim 10^{-16}$. For a Lorentzian pedestal of half-width $\Gamma = c/\ell_P \approx 1.9\times10^{43}\,$Hz, the in-band kernel fraction over the LHAASO acceptance (3–13 TeV, $\Delta\nu \approx 2.4\times10^{27}\,$Hz) is $f_{\rm band} = \Delta\nu/(\pi\Gamma) \approx 4\times10^{-17}$, so scattered flux returning to band is negligible.

With the in-band pedestal negligible, the observed flux reduces to the *absolute* suppression

$$S_{\rm obs}(\nu)\Big|_{\text{in-band}} \;\approx\; e^{-\sigma_\phi^2(\nu)}\, S_{\rm src}(\nu). \tag{22}$$

This is an absolute attenuation, testable whenever the detected flux can be compared to an independently predicted intrinsic flux. Equation (22) holds while the surviving core exceeds the pedestal floor, $\sigma_\phi^2 \lesssim -\ln f_{\rm band} \approx 38$ (Section IV); for stronger foam the in-band signal saturates at the floor $\sim f_{\rm band} N_0$, an escape foreclosed below by the transient, directional nature of the detection.

The LHAASO-KM2A detector registered 142 gamma rays above 3 TeV from GRB 221009A at $20.6\,\sigma$ significance, with events extending to $\sim$13 TeV [18, 19]. By Eq. (22) the observed in-band count is $N \approx e^{-\sigma_\phi^2(E_{\rm TeV})}\, N_0$, where N_0 is the in-band count in the absence of foam (after absorption by the extragalactic background light, EBL). Detection of N photons implies

$$\sigma_\phi^2(E_{\rm TeV}) \;\approx\; \ln\frac{N_0}{N} \;\equiv\; \sigma^2_{\phi,\max}, \tag{23}$$

where $\sigma^2_{\phi,\max}$ is the number of e-foldings of suppression the source can sustain before the foam-attenuated count falls below the observed N. Writing $\sigma_\phi^2 = (2\pi D_C/\lambda_{\rm obs})^2(\ell_P/D_C)^{2\alpha}$ from Eqs. (1) and (4) ($A_\alpha = 1$, $\lambda_{\rm obs} = hc/E_{\rm TeV}$) and inverting gives

$$\alpha \;\gtrsim\; \frac{\ln\!\big(2\pi D_C/\lambda_{\rm obs}\big) - \tfrac{1}{2}\ln\sigma^2_{\phi,\max}}{\ln(D_C/\ell_P)} \;\equiv\; \alpha_{\rm TeV}. \tag{24}$$

The strength of this bound is best seen by reductio. At the holographic value $\alpha = 2/3$, Eqs. (1)–(4) at $E_{\rm TeV} = 3\,$TeV give $\sigma_\phi^2 \approx 7\times10^8$. To produce the observed 142 photons would therefore require an intrinsic in-band count $N_0 = N\, e^{\sigma_\phi^2} \approx 142\, e^{7\times10^8} \approx 10^{3\times10^8}$ — exceeding any astrophysical source by hundreds of millions of orders of magnitude. The exclusion is immune to any uncertainty in the intrinsic spectrum or EBL model.

The precise boundary depends on how many e-foldings the source can tolerate: allowing $\sigma^2_{\phi,\max} = 1$, 3, or 10 e-foldings gives $\alpha_{\rm TeV} = 0.740$, 0.736, or 0.732 respectively ($E_{\rm TeV} = 3\,$TeV, $\ln(2\pi D_C/\lambda_{\rm obs}) \approx 102.4$, $\ln(D_C/\ell_P) \approx 138.4$). At the highest detected energy $E_{\rm TeV} \approx 13\,$TeV the bound tightens to $\alpha \gtrsim 0.75$. The result is doubly logarithmic in

N_0/N: a decade in the tolerated suppression shifts α by less than 0.01 (at $z = 0.151$ the EBL optical depth at 3 TeV is only $\tau_{\rm EBL} \sim 0.3$).

The escape could instead lie in the pedestal kernel itself — a light-tailed $\hat{R}$ that, contrary to Assumption (ii), returns a non-negligible fraction of the redistributed flux to the observing band. The transient, directional nature of the detection forecloses this: the kernel acts on every source at all times, so appreciable band-refilling would deposit flux as a persistent sky-wide background — not as a $20.6\,\sigma$ excess from the direction of GRB 221009A, coincident with the burst. The 142 TeV photons are therefore genuine survivors of the source's beamed emission, and the suppression they constrain is real. The same reasoning bounds the kernel at the population level: a refilling kernel would reprocess the entire steady γ-ray source population into a diffuse glow far exceeding the observed background, an overproduction sharpest in the γ-ray band itself because $\sigma_\phi^2 \propto \nu^2$ concentrates the reprocessing at high energies. That detecting the suppression requires a once-in-10^4-year transient, rather than a uniformly glowing sky, is itself the statement that the in-band refill is negligible.

3. Energy-dependent line width

The line-width probe is complementary because it is *relative*: the absolute suppression $e^{-\sigma_\phi^2(\nu)}$ attenuates line peak and adjacent continuum identically, canceling in the fractional width σ/E. It therefore needs no absolute-flux model — foam is read off the line shape alone — and is *pedestal-independent*, immune to the redistribution physics contested above, surviving rejection of Assumption (ii), the only escape from the TeV bound. Its limitation is that, for a narrow line, the fractional width responds to foam only at second order in the jitter (Eq. (27) below), so it constrains α only in the strong-foam regime and yields a significantly weaker, if independent, floor on α.

a. The intrinsic line as a Doppler invariant. The MeV feature is understood as a Doppler-shifted line from the relativistic outflow — most plausibly the blueshifted 511 keV electron–positron annihilation line of pair-plasma clumps comoving with the jet [11, 15], though what follows needs only that it is a boosted line, not the specific transition. Its observed centroid is then the comoving line blueshifted by the bulk Doppler factor $\mathcal{D}$ of the emitting material, $E_{\rm line} = \mathcal{D}\, m_e c^2/(1+z)$, so the centroid measures $\mathcal{D}$ directly; the observed decline $E_{\rm line} \propto t^{-1}$ from ≈ 37 to ≈ 6 MeV tracks the falling jet Lorentz factor ($\Gamma \propto t^{-1}$, $\Gamma \approx 84$ at 37 MeV in the Zhang et al. modeling) as the outflow evolves [15]. Because a bulk boost multiplies every photon energy — line center and line width alike — by the same $\mathcal{D}$, the *fractional* width $r_{\rm int} = \sigma'/E'$ is boost-invariant: it equals its comoving value at every epoch, independent of $\mathcal{D}$, and is therefore expected to stay constant as the burst evolves and the centroid sweeps downward. This is what is observed — $r_{\rm obs} = 0.10 \pm 0.01$ across the seven sub-intervals (Section V A 1) — so in the conventional picture $r_{\rm int}$ is a fixed, epoch-independent property of the source, set by the comoving line shape (the spread of clump Doppler factors and the pair-plasma temperature [15]) and carrying no foam. Quantum-gravity jitter, if present, imprints an *additional*, energy-dependent broadening on top of this constant baseline, computed next.

b. Derivation of fractional-width broadening. Eq. (17) gives an observed core whose absolute width $\sigma_{\rm obs} = 1/[2\pi\sqrt{\tau_G^2 + \sigma_\Delta^2}]$ is *narrower* than the intrinsic $\sigma_\nu = 1/(2\pi\tau_G)$ (item (ii), Section III B); however, the line center shifts down to $\nu_0^{\rm eff} = \nu_0 \tau_G^2/(\tau_G^2 + \sigma_\Delta^2)$ (item (i)) by a larger relative amount. The *fractional* width is $r_{\rm obs} \equiv \sigma_{\rm obs}/\nu_0^{\rm eff}$, with $\nu_0^{\rm eff}$ the observed centroid. Combining gives

$$r_{\rm obs} = \frac{\sigma_{\rm obs}}{\nu_0^{\rm eff}} = \frac{1/(2\pi\sqrt{\tau_G^2+\sigma_\Delta^2})}{\nu_0\,\tau_G^2/(\tau_G^2+\sigma_\Delta^2)} = \frac{1}{2\pi\nu_0\tau_G}\,\frac{\sqrt{\tau_G^2+\sigma_\Delta^2}}{\tau_G} = r_{\rm int}\sqrt{1+\sigma_\Delta^2/\tau_G^2}, \tag{25}$$

so the fractional width *broadens*:

$$r_{\rm obs}^2(E) \;=\; r_{\rm int}^2\big[1+\sigma_\Delta^2/\tau_G^2(E)\big] \;=\; r_{\rm int}^2 + r_{\rm foam}^2(E), \tag{26}$$

with $r_{\rm foam}^2(E) \equiv r_{\rm int}^2\,\sigma_\Delta^2/\tau_G^2(E)$.

In the dynamical limit $\sigma_\Delta = \sqrt{2}\,\sigma_\ell/c$ saturates, and with $\tau_G(E) = \lambda_{\rm obs}(E)/(2\pi c\, r_{\rm int})$ the foam excess becomes

$$r_{\rm foam}(E) \;=\; \sqrt{2}\,r_{\rm int}^2\,\sigma_\phi(E) \;=\; r_{\rm int}^2\cdot\frac{2\sqrt{2}\,\pi\,\sigma_\ell}{\lambda_{\rm obs}(E)} \;\propto\; E, \tag{27}$$

predicting chromatic broadening that grows linearly with photon energy. The $r_{\rm int}^2$ prefactor makes the fractional width a *second-order* foam probe: since $\sigma_\Delta^2/\tau_G^2 = 2r_{\rm int}^2\sigma_\phi^2$, an order-unity broadening of a narrow line ($r_{\rm int} \approx 0.1$) requires $\sigma_\phi^2 \gtrsim 40$.

c. Fit and bound. A χ^2 fit of the seven measurements to the two-parameter model

$$r^2_{\rm obs}(E) \;=\; r^2_{\rm int} + r^2_{\rm foam,0}\,(E/E_0)^2, \qquad E_0 = 37\,{\rm MeV}, \tag{28}$$

constrains the foam amplitude. The best fit lies at the boundary $r_{\rm foam,0} = 0$ (with $r_{\rm int} = 0.111$, $\chi^2_{\rm min} = 5.3/6\,{\rm dof}$), confirming the constant-$r_{\rm obs}$ hypothesis. Profiling over $r_{\rm int}$ gives the 95% CL upper limit $r^{\rm max}_{\rm foam,0} = 0.27$ at $\Delta\chi^2 = 2.71$ (boundary-corrected Chernoff threshold [20]: $\frac{1}{2}\delta_0 + \frac{1}{2}\chi^2_1$ for zero-bounded parameters; immaterial to α given the logarithmic lever arm below). Converting via Eq. (27) and $\sigma^2_\ell = D_C^{2(1-\alpha)}\ell_P^{2\alpha}$ (Eq. (1), $A_\alpha = 1$) gives

$$\alpha \;\gtrsim\; \frac{\ln(D_C/\lambda_{\rm obs}) + |\ln r^{\rm max}_{\rm foam,0}| - |\ln(2\sqrt{2}\,\pi\, r^2_{\rm int})|}{\ln(D_C/\ell_P)} \;=\; \frac{89.3 + 1.3 - 3.3}{138.4} \;\approx\; 0.63\,. \tag{29}$$

Here $r_{\rm int} = 0.066$ is the width profiled jointly with $r^{\rm max}_{\rm foam,0}$ on the 95% boundary (not the global best fit 0.111), giving $|\ln(2\sqrt{2}\,\pi\, r^2_{\rm int})| \approx 3.3$. Like the TeV bound, the result is doubly logarithmic and so dominated by the kinematic lever arm $\ln(D_C/\ell_P) = 138$: a tenfold change in $r^{\rm max}_{\rm foam,0}$ moves α by only $\ln 10/\ln(D_C/\ell_P) \approx 0.017$, so $\alpha \gtrsim 0.63$ is fixed by the geometry, not by the precision of the seven measurements.

d. Validity and caveats. At the 95% boundary $\sigma_\Delta/\tau_G(E_0) \approx 4.1$, i.e. $\sigma^2_\phi(37\,{\rm MeV}) \sim 10^3$, the bound lies deep in the strong-foam regime, and three caveats attach to it. First, the fractional-width law Eq. (26) is a property of the coherent core, whose weight $e^{-\sigma^2_\phi}$ is exponentially small there; the bound thus extrapolates the core's width relation into a regime where the line flux has largely been redistributed into the pedestal, and is best read as an order-of-magnitude exclusion anchored by the logarithmic lever arm rather than a sharp threshold. Second, the same Eq. (15) predicts a centroid depression $\nu_0^{\rm eff}/\nu_0 = \tau_G^2/(\tau_G^2 + \sigma_\Delta^2) \approx 0.06$ at 37 MeV at this boundary, absent from the smooth $E_{\rm line} \propto t^{-1}$ trend in the data; the width-only fit treats observed centroids as intrinsic, and a joint centroid-plus-width analysis is left to future work. Third, the bound rests on the seven sub-interval widths of the single GRB 221009A feature, whose most constraining energy (37 MeV) lies in the Fermi MeV sensitivity gap (Section VI), and the constant-$r_{\rm int}$ fit cannot distinguish a genuine null from small epoch-to-epoch variation of the intrinsic width. With these caveats, the width-only $\alpha \gtrsim 0.63$ stands as the pedestal-independent floor beneath the TeV bound.

B. CMB

The CMB blackbody is itself a broad, asymmetric spectral profile, so foam acts on it exactly as on any line: the coherent core is suppressed by $e^{-\sigma^2_\phi(\nu)}$ and the removed flux reappears as a broad pedestal whose shape depends on the short-time behavior of $C_\alpha(\tau)$. Because $\sigma^2_\phi \propto \nu^2$ (Eq. (4)), the foam deficit is concentrated at high frequencies — a ν^2-weighted suppression distinct from the standard chemical-potential (μ) and Compton-y distortions. For Planck-scale foam the pedestal is so broad that it lies almost entirely outside the observing band (Section V B 2), leaving an essentially pure deficit in the FIRAS window.

The Far-InfraRed Absolute Spectrophotometer (FIRAS) aboard the Cosmic Background Explorer (COBE) measured the absolute spectrum of the CMB monopole over 60–600 GHz by direct comparison against an internal cryogenic blackbody calibrator — the absolute-photometry feature that distinguishes it from relative-photometry experiments and makes it well suited to constraining spectral *shape* distortions. The calibrated monopole agrees with a pure blackbody to within ~ 50 ppm (a few $\times 10^{-5}$ fractional) across the band, placing the still-standard limits $|\mu| \lesssim 9 \times 10^{-5}$ and $|y| \lesssim 1.5 \times 10^{-5}$ [21]. These limits and the 43-channel calibrated spectrum form the empirical basis for the foam constraint derived below.

1. Foam phase variance at CMB frequencies

CMB photons travel from the last-scattering surface at comoving distance $D_{\rm C} = 13.87$ Gpc (Planck 2018 flat-ΛCDM cosmology [17]; $H_0 = 67.36\,{\rm km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$). The foam phase variance (Eq. (4)) is set by the effective foam amplitude $A_{\rm eff} = A_\alpha D_{\rm C}^{2(1-\alpha)}\ell_P^{2\alpha}$ (the lumped amplitude entering the FIRAS fit, Section V B 3):

$$\sigma^2_\phi(\nu) = \left(\frac{2\pi\nu}{c}\right)^2 A_{\rm eff}, \tag{30}$$

with $A_\alpha = 1$.

At $\nu = 150\,\mathrm{GHz}$ this gives

$$\sigma_\phi^2(150\,\mathrm{GHz}) \approx \begin{cases} 7\times10^{-2} & \alpha = 1/2 \text{ (random walk)}, \\ 2\times10^{-22} & \alpha = 2/3 \text{ (holographic)}. \end{cases} \tag{31}$$

For $\alpha = 1/2$ the variance reaches order unity at the top of the FIRAS band ($\sigma_\phi^2 \approx 1.2$ at 639 GHz), and the linearised template holds for $\alpha \gtrsim 0.50$, where $\sigma_\phi^2(\nu) \lesssim 1$ through the band (Section V B 3).

2. Foam-modified CMB spectrum

Given a foam model, the natural analysis is of the whole modified spectrum: compute $g_{\mathrm{src}}^{(1)}(\tau)$ as the Fourier transform of $B_\nu(T_0)$ (Eq. (16)), apply the jitter convolution (11), and inverse-transform to $S_{\mathrm{obs}}(\nu)$. We instead separate it into a coherent core and a pedestal, because the split isolates a *model-independent* piece — the suppression $e^{-\sigma_\phi^2(\nu)}$, exact for any C_α (Section IV) — from the *model-dependent* pedestal kernel $\hat{R}(\nu-\nu';\nu')$, a decomposition that yields general physical insight the combined spectrum obscures. A worked example with exponential $C_\alpha(\tau)$ below identifies the physical no-pedestal case used in Section V B 3.

Decomposing the blackbody as an incoherent superposition of narrowband modes $S_{\mathrm{src}}(\nu') = B_{\nu'}(T_0)$, each obeys the *same* achromatic jitter with per-mode foam factor

$$\mathcal{F}_{\mathrm{foam}}(\tau;\nu') = e^{-\sigma_\phi^2(\nu')[1-C_\alpha(\tau)]}, \quad \sigma_\phi^2(\nu') = (2\pi\nu'/c)^2\sigma_\ell^2(D), \tag{32}$$

with $\sigma_\ell^2(D) = A_{\mathrm{eff}}$ (Section V B 1). By the core-plus-pedestal split of Section IV, in the dynamical limit each mode's long-lag plateau $e^{-\sigma_\phi^2(\nu')}$ Fourier-transforms to a coherent core at ν' and its short-lag residual to a broad pedestal; superposing over the Planck modes (intensity-weighted Eq. (16)) gives

$$S_{\mathrm{obs}}(\nu) = e^{-\sigma_\phi^2(\nu)}\,B_\nu(T_0) + \int_0^\infty B_{\nu'}(T_0)\,\hat{R}(\nu-\nu';\nu')\,d\nu', \tag{33}$$

with $\hat{R}$ the Fourier transform of the residual, carrying the full C_α-dependence. By the unitarity of the per-mode modulation $e^{i\omega\delta t(t)}$, the pedestal conserves photon number, $\int[S_{\mathrm{obs}}(\nu) - B_\nu(T_0)]\,d\nu/(h\nu) = 0$.

To make the pedestal kernel concrete, we evaluate $\hat{R}$ for the minimal model $C_\alpha(\tau) = e^{-|\tau|/\tau_{\mathrm{foam}}}$ (a Markovian metric fluctuation). Substituting into Eq. (7) gives

$$\sigma_\Delta^2(\tau) = \frac{2\sigma_\ell^2(D)}{c^2}\left[1 - e^{-|\tau|/\tau_{\mathrm{foam}}}\right]. \tag{34}$$

The Fisher analysis below operates in the linearised regime $\sigma_\phi^2(\nu') \ll 1$. Expanding $\mathcal{F}_{\mathrm{foam}}(\tau;\nu')$ (Eq. (32)) to first order and subtracting the long-lag asymptote $e^{-\sigma_\phi^2(\nu')}$ leaves the short-τ residual

$$\mathcal{F}_{\mathrm{foam}}(\tau;\nu') - e^{-\sigma_\phi^2(\nu')} \;\approx\; \sigma_\phi^2(\nu')\,e^{-|\tau|/\tau_{\mathrm{foam}}}, \tag{35}$$

whose Fourier transform is a Lorentzian centered on the source frequency,

$$\hat{R}(\nu-\nu';\nu') \;=\; \sigma_\phi^2(\nu')\,\frac{2\tau_{\mathrm{foam}}}{1 + [2\pi(\nu-\nu')\tau_{\mathrm{foam}}]^2}, \tag{36}$$

of full width at half maximum $1/(\pi\tau_{\mathrm{foam}})$. For Planck-scale foam with $\tau_{\mathrm{foam}} \sim \ell_P/c \approx 5\times10^{-44}\,\mathrm{s}$, the Lorentzian width ($\sim 10^{43}\,\mathrm{Hz}$) lies thirty-one orders of magnitude beyond the FIRAS upper edge ($\nu_{\max} = 6\times10^{11}\,\mathrm{Hz}$), so the in-band flux fraction is $\sim 10^{-31}$. The physical pedestal thus deposits no detectable in-band power for any plausible τ_{foam}. The constraint of Section V B 3 consequently adopts the no-pedestal limit as the primary template.

3. Constraint from FIRAS

We use $I_\nu \equiv S_{\mathrm{obs}}(\nu)$ to denote the observed CMB intensity including the foam signal. Working in the linearised regime $\sigma_\phi^2(\nu) \ll 1$, we treat four spectral parameters $\boldsymbol{\theta} = (A_{\mathrm{eff}}, T_0, \mu, y)$, where $A_{\mathrm{eff}} = A_\alpha D_C^{2(1-\alpha)}\ell_P^{2\alpha}$ is the cumulative

comoving foam variance at the CMB distance (Section V B 1), so that $\sigma_\phi^2(\nu) = (2\pi\nu/c)^2 A_{\rm eff}$. Note that α is not fitted directly; only the lumped amplitude $A_{\rm eff}$ enters the fit, and bounds on α are obtained *a posteriori* by inverting the Ng–Perlman relation at fixed $A_\alpha = 1$.

For physical foam the pedestal contributes no in-band power (Section V B 2), so the only in-band foam signature is the pure deficit — the no-pedestal limit of Eq. (33) —

$$\frac{\partial I_\nu}{\partial A_{\rm eff}} = -\left(\frac{2\pi\nu}{c}\right)^2 B_\nu(T_0), \tag{37}$$

a ν^2-weighted (blue-tilted) Planck spectrum. The chemical-potential and Compton-y distortion templates are

$$\frac{\partial I_\nu}{\partial \mu} = -B_\nu(T_0)\,\frac{x e^x}{(e^x-1)^2}, \tag{38}$$

$$\frac{\partial I_\nu}{\partial y} = \quad B_\nu(T_0)\,\frac{x e^x}{e^x-1}\left[\frac{x(e^x+1)}{e^x-1} - 4\right], \tag{39}$$

with $x = h\nu/(k_B T_0)$, encoding respectively early-universe energy injection (Silk damping, decaying particles) and late-time electron heating (reionization, cluster gas).

We constrain $A_{\rm eff}$ by fitting these templates — together with the FIRAS Galactic-foreground template — to the public COBE/FIRAS monopole residual spectrum and its calibration covariance [21]. Raising T_0 brightens the Planck curve broadband; the foam deficit suppresses it with an extra ν^2 tilt. Over the CMB-bright band (peak through the onset of the Wien fall-off), the two differ only by this tilt, making the foam template nearly indistinguishable from a temperature shift ($\rho_{A_{\rm eff} T_0} \simeq 1$ in the fit) and thus strongly collinear with the y and μ templates (forecast correlation matrix Eq. (41) below). Using the full 43×43 apodization covariance, the generalized least-squares fit over $(T_0, \mu, y, A_{\rm eff})$ and the foreground amplitude recovers μ and y consistent with zero (foam-marginalized 95% limits $|\mu| < 1.7 \times 10^{-4}$, $|y| < 1.0 \times 10^{-5}$; $\chi^2/\nu = 1.24$ — the μ limit looser than the foam-free Fixsen value $|\mu| \lesssim 9 \times 10^{-5}$ because marginalizing over the foam amplitude inflates $\sigma(\mu)$, as quantified below) and a foam amplitude consistent with zero,

$$A_{\rm eff} = (-2.4 \pm 5.3) \times 10^{-11}\,{\rm m}^2. \tag{40}$$

Inverting $A_{\rm eff} = A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}$ at $A_\alpha = 1$, the one-sided 95% upper limit $A_{\rm eff} < 1.645\,\sigma(A_{\rm eff}) = 8.7 \times 10^{-11}\,{\rm m}^2$ gives the lower bound $\alpha \gtrsim 0.52$ (collected with the forecasts below in Table I). This bound is degeneracy-limited: $\alpha \gtrsim 0.52$ rests only on the residual spectral-tilt information that distinguishes the ν^2-weighted deficit from a temperature shift, so marginalizing over (T_0, μ, y) leaves it weak. The only direction that could break the foam–T_0 degeneracy is deep into the Wien/sub-mm regime, where the ν^2-tilt separates from a temperature shift; but there, beyond the FIRAS band, galactic dust and the cosmic infrared background dominate, making the monopole impractical. The fit uses the linearised template $\Delta I_\nu \approx \sigma_\phi^2(\nu)$ [template] (valid for $\sigma_\phi^2 \ll 1$), which holds across the FIRAS band at the headline threshold: σ_ϕ^2 exceeds unity only for stronger, excluded foam ($\alpha \lesssim 0.50$; Section V B 1). Replacing the linearised deficit $-\sigma_\phi^2 B_\nu$ by the exact form $-(1 - e^{-\sigma_\phi^2}) B_\nu$ (valid at all σ_ϕ^2, Section IV) shifts the integrated in-band amplitude by $\approx 13\%$ at $\alpha = 0.50$ but by $< 1\%$ at $\alpha \gtrsim 0.52$ ($\Delta\alpha \sim 3 \times 10^{-5}$), so the bound is unaffected; in that strong-foam regime nonlinearity may even help break the foam–μ degeneracy.

The foam–T_0 degeneracy has two consequences for the CMB temperature itself. First, admitting foam as a free parameter inflates the marginalized temperature uncertainty: the FIRAS spectral-shape determination of T_0 degrades from $1.2 \times 10^{-5}\,{\rm K}$ to $7.7 \times 10^{-4}\,{\rm K}$ (a factor ~ 64; $\sigma(y)$ and $\sigma(\mu)$ inflate by ~ 17 and ~ 3), comparable to the full COBE/FIRAS calibration uncertainty $5.7 \times 10^{-4}\,{\rm K}$ [22]. A foam-agnostic monopole analysis thus roughly doubles the temperature error budget, and the quoted precision of $T_0 = 2.72548\,{\rm K}$ implicitly assumes no foam. Second, foam present but *omitted* from the fit shifts the central value by $\Delta T_0 = b_{T_0} A_{\rm eff}$ with $b_{T_0} \approx -1.5 \times 10^7\,{\rm K\,m^{-2}}$; for $A_\alpha = 1$ this is only $\approx 3 \times 10^{-22}\,{\rm K}$ at the holographic $\alpha = 2/3$ and vanishes as $\alpha \to 1$ (where the comoving foam variance ceases to grow with distance), reaching $\sigma(T_0)$ only at $\alpha \approx 0.52$ where the foam is already a many-σ signal. Both effects therefore matter only for foam near FIRAS detectability; the recovered T_0 is immune to any foam consistent with the GRB bound $\alpha \gtrsim 0.63$.

4. Fisher forecasts for future spectrometers

To forecast future absolute spectrometers we propagate the no-pedestal templates through a Fisher matrix calibrated to FIRAS ($\sigma_\nu = 0.01\,{\rm MJy/sr}$). The forecast gives $\sigma(A_{\rm eff}) = 2.2 \times 10^{-11}\,{\rm m}^2$ and $\alpha \gtrsim 0.52$ (reproducing the data fit);

the conditional value $\sigma(A_{\rm eff})|_{\mu,y\,\rm fixed} = 8.2\times10^{-13}\,\mathrm{m}^2$ shows that marginalizing over distortions costs a factor ~ 27 via the foam–T_0 collinearity. The fiducial degeneracy structure is captured by the correlation matrix for ($A_{\rm eff}$, T_0, μ, y),

$$\rho_{ij}^{\rm fid} = \begin{pmatrix} 1.000 & \mathbf{0.9997} & 0.928 & 0.999 \\ \mathbf{0.9997} & 1.000 & 0.934 & 0.998 \\ 0.928 & 0.934 & 1.000 & 0.915 \\ 0.999 & 0.998 & 0.915 & 1.000 \end{pmatrix}, \tag{41}$$

the foam deficit is almost perfectly aligned with a temperature shift ($\rho_{A_{\rm eff}T_0} = 0.9997$, bold) and strongly collinear with the y and μ distortions (0.999 and 0.928). Defining the marginal detection SNR as $A_{\rm eff}/\sigma(A_{\rm eff})$, so that SNR $= 1.645$ marks the one-sided 95% CL threshold, Figure 1 plots it versus α, and Table I collects the thresholds.

TABLE I. Marginal foam-detection thresholds (one-sided 95% CL). *Lower* bounds on α ($A_\alpha = 1$): the foam signal $\sigma_\ell^2 \propto (\ell_P/D_C)^{2\alpha}$ decreases with α, so non-detection excludes small α. FIRAS: data fit (Section V B 3). PIXIE, SuperPIXIE, Voyage 2050: Fisher forecast scaled to published μ-sensitivities [23]. The three forecast entries (0.545, 0.549, 0.554) round to 0.54–0.55.

	lower bound
FIRAS	$\alpha \gtrsim 0.52$
PIXIE	$\alpha \gtrsim 0.54$
SuperPIXIE	$\alpha \gtrsim 0.55$
Voyage 2050	$\alpha \gtrsim 0.55$

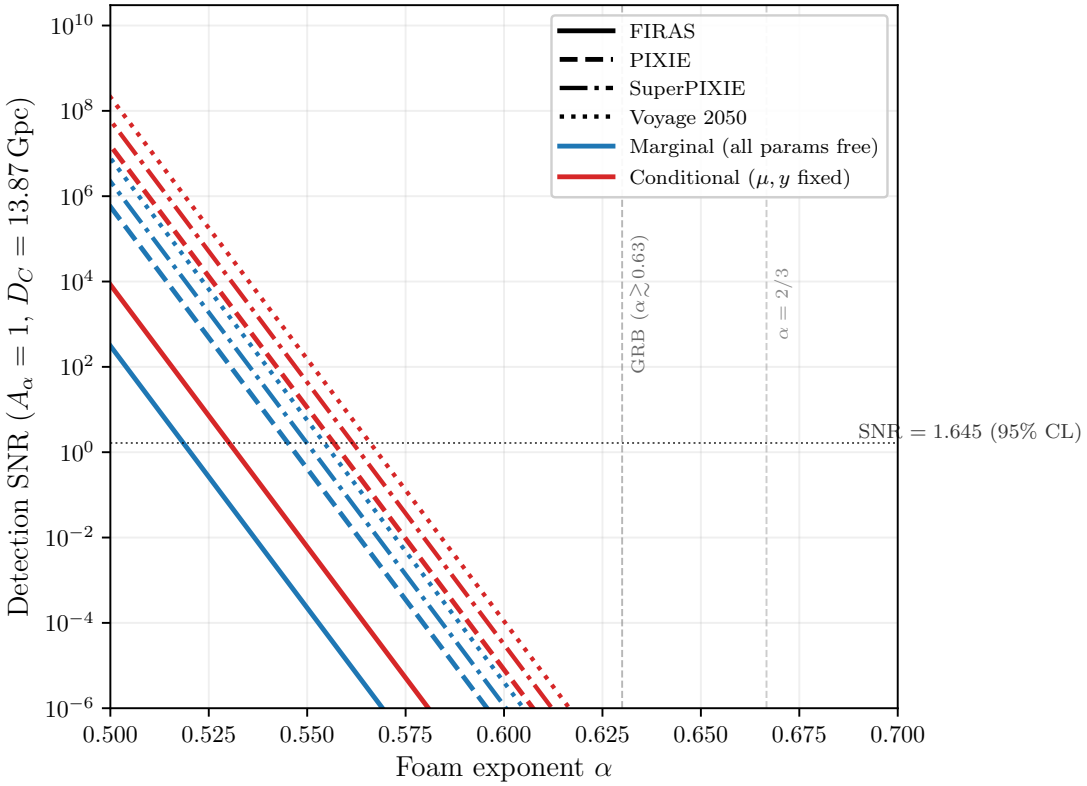


FIG. 1. Detection SNR for the foam amplitude $A_{\rm eff}$ versus exponent α ($A_\alpha = 1$, $D_C = 13.87\,$Gpc). Blue: marginal SNR (all four parameters free). Red: conditional SNR (μ, y fixed), $\sim 27\times$ higher. Line styles: solid FIRAS, dashed PIXIE, dash-dotted SuperPIXIE, dotted Voyage 2050 (forecast μ-sensitivities scaled to FIRAS; [23]). The marginal lower bounds cluster at $\alpha \approx 0.52$–0.55. Horizontal dotted line: SNR $= 1.645$ (one-sided 95% CL). Vertical dashed lines: the spectroscopic lower bound from GRB 221009A ($\alpha \gtrsim 0.63$) and the holographic prediction $\alpha = 2/3$.

As a worst case, we test an *adversarial* in-band pedestal shaped like the μ-distortion template (Rayleigh–Jeans

$1/\nu$), maximizing the foam–μ overlap. The Fisher analysis rotates the foam template off the temperature axis ($\rho_{A_{\rm eff}T_0} : 0.9997 \to 0.937$) and off the y template ($0.999 \to 0.010$), leaving a foam–μ correlation of 0.976. Because $0.976 < 0.9997$, the amplitude is *better* constrained and the threshold *tightens* to $\alpha \gtrsim 0.53$ ($\Delta\alpha \approx 0.009$ for the factor $N \approx 13$ in amplitude sensitivity), so an in-band pedestal cannot loosen the bound. An external μ prior (e.g. the ΛCDM dissipation value $\mu \approx 2 \times 10^{-8}$ [23]) sharpens $\sigma(A_{\rm eff})$ by ~ 3–$5\times$ in this μ-limited adversarial case but barely helps the physical no-pedestal case, whose limiting degeneracy is with T_0; either way the logarithmic relation caps the gain at $\Delta\alpha \lesssim 0.005$. The physical no-pedestal limit is therefore the conservative case and remains the headline; both kernels are chosen probes, but the in-band pedestal is negligible for any C_α with $\tau_{\rm foam} \ll 1/\nu_{\rm max}$ (Section V B 2).

The proposed PIXIE satellite [24] — a fully cryogenic ($T_{\rm sys} \approx 2.7\,{\rm K}$) nulling polarizing Fourier transform spectrometer spanning 30 GHz to 6 THz — is projected to surpass FIRAS sensitivity by roughly three orders of magnitude, reaching $|\mu| \lesssim 10^{-8}$ and $|y| \lesssim 10^{-9}$ and entering the regime of guaranteed ΛCDM spectral signals from Silk damping and reionization. Propagating this gain through the Fisher-matrix scaling derived above, the marginal foam detection threshold shifts from $\alpha \gtrsim 0.52$ (FIRAS) to approximately $\alpha \gtrsim 0.54$ (PIXIE; Table I), narrowing the gap with the spectroscopic lower bound $\alpha \gtrsim 0.63$ from GRB 221009A (Fig. 1). All thresholds are quoted for the physical no-pedestal deficit. Step sizes track the logarithm of the sensitivity ratio: each decade of $\sigma(\mu)$ buys $\Delta\alpha = \ln 10/[2\ln(D_C/\ell_P)] \approx 0.008$. The $\sim$3-decade FIRAS→PIXIE gain gives the ≈ 0.03 step; PIXIE→Voyage 2050 (under a decade) adds ≈ 0.009. The thresholds saturate monotonically (the three forecast entries 0.545, 0.549, 0.554 round to 0.54–0.55). The foam–T_0 degeneracy persists on the linearized template. But PIXIE's extension to 6 THz enters the nonlinear regime $\sigma_\phi^2 \gg 1$ for $\alpha \lesssim 0.52$ (higher than the FIRAS crossing $\alpha \lesssim 0.50$ because the wider band pushes the $\sigma_\phi^2 = 1$ boundary upward; Section V B 3), where the foam deficit diverges markedly from the blackbody and μ-distortion shapes, possibly reducing the degeneracy. The $\alpha \gtrsim 0.54$ projection may therefore understate the reach: if the nonlinear template effectively breaks this degeneracy at PIXIE sensitivity, the true threshold could be significantly more stringent.

Extending the same Fisher analysis to the 10 GHz–3 THz bands of the SuperPIXIE and Voyage 2050 concepts [23] (forecast distortion sensitivities $\sigma(\mu) \approx 7.7 \times 10^{-9}$ and 2.2×10^{-9}) shifts the marginal threshold only to $\alpha \gtrsim 0.55$ (Table I), barely beyond the PIXIE projection. The reason is structural rather than instrumental: in the no-pedestal limit the foam–T_0 degeneracy ($\rho^{\rm fid}_{A_{\rm eff}T_0} = 0.9997$) is *band-robust*: broadening coverage from the FIRAS band to 10 GHz–6 THz leaves it unchanged to four decimal places. The wider band does break the foam–μ degeneracy probed by the μ-aligned pedestal (from $\rho_{A_{\rm eff}\mu} = 0.98$ at FIRAS to 0.66 over 10 GHz–3 THz), so future missions are markedly more robust against in-band pedestal contamination, but, within the linearized analysis, the physical threshold does not move (the nonlinear, strong-foam exception noted above, $\alpha \lesssim 0.52$, aside). With the logarithmic Ng–Perlman scaling $\Delta\alpha = \ln N/[2\ln(D_C/\ell_P)]$ converting even a 10^4-fold sensitivity gain over FIRAS into $\Delta\alpha \lesssim 0.04$, this sets a firm ceiling: as a broadband probe based on absolute calibrated photometry (no source-profile assumption, a different systematic-error budget from spectroscopy), spectral-distortion photometry yields $\alpha \gtrsim 0.52$ from FIRAS and at most $\alpha \gtrsim 0.55$ next-generation, the bound limited by the foam–T_0 degeneracy rather than by sensitivity or bandwidth.

VI. CONCLUSIONS

Spacetime foam imprints an achromatic, stochastic arrival-time jitter that modifies the first-order temporal coherence $g^{(1)}(\tau)$, and hence the observed spectrum, providing a spectroscopic probe complementary to the imaging (PSF, Strehl) channel. In the dynamical-foam limit the spectrum splits into a coherent line core suppressed by $e^{-\sigma_\phi^2}$ and an incoherent pedestal, yielding a shape-distortion bound on the Ng–Perlman exponent α. The MeV emission lines of GRB 221009A give $\alpha \gtrsim 0.63$ (one-sided 95% CL), while a fit to the COBE/FIRAS monopole gives $\alpha \gtrsim 0.52$ (95% CL); next-generation spectrometers (PIXIE, SuperPIXIE, Voyage 2050) reach $\alpha \gtrsim 0.54$–0.55, the bound saturating logarithmically because the foam deficit is nearly degenerate with a temperature shift. The pedestal-independent linewidth and CMB bounds do not exclude the holographic value $\alpha = 2/3$; however, the TeV survival bound ($\alpha_{\rm TeV} \gtrsim 0.74$, conditional on the pedestal-kernel shape of Assumption (ii)) would exclude it. Probing the same exponent through different sources, observables, and systematics, the two channels form an independent cross-check rather than a joint bracket. They are separated by more than ten orders of magnitude in frequency (CMB photons near 10^{11} Hz versus the GRB's MeV–TeV emission at 10^{22}–10^{27} Hz), and since the foam phase variance grows as $\sigma_\phi^2 \propto \nu^2$, this lever arm makes the high-frequency GRB channel intrinsically more sensitive, hence the more constraining. This sensitivity advantage is partly offset by a firmer baseline on the CMB side: its intrinsic spectrum is a near-perfect blackbody, whereas the GRB bound depends on a less certain model of the source's intrinsic emission. Probing the same exponent through different sources, observables, and systematics, the two channels form an independent cross-check rather than a joint bracket. They are separated by more than ten orders of magnitude in frequency (CMB photons near 10^{11} Hz versus the GRB's MeV–TeV emission at 10^{22}–10^{27} Hz), and since the foam phase variance grows as $\sigma_\phi^2 \propto \nu^2$, this lever arm makes the high-frequency GRB channel intrinsically more sensitive, hence the more constraining.

All single-source spectroscopic bounds derived in this paper assume the dynamical-foam limit $\tau_{\rm foam} \ll \tau_c$ (Section III B), in which the foam path-length jitter decorrelates many times within each photon coherence time. If foam fluctuations are instead slow ($\tau_{\rm foam} \gg \tau_c$), the accumulated phase is essentially frozen during each coherence time and the foam acts as a deterministic, frequency-independent phase screen — no spectral broadening or peak reduction occurs within a single observation, and the shape-distortion bounds do not apply. The CMB constraint (Section V B) is similarly derived in the dynamical limit (Section V B 2 works in $C_\alpha \to 0$): in the opposite quasi-static limit the per-mode foam suppression $\mathcal{F}_{\rm foam}(\tau) \to 1$ (Eq. (13)) and the CMB monopole is unmodified, so the shape-distortion bounds do not apply in either channel.

The pedestal-independent dynamical bound $\alpha \gtrsim 0.63$ (Eq. (29)) — a weak, lever-arm-dominated floor — does not exclude the holographic value $\alpha = 2/3 \approx 0.667$, though the TeV bound (Eq. (24), conditional on Assumption (ii)) would. A principal instrumental limitation of the present analysis is the MeV sensitivity gap: the spectral region between the upper edge of the Fermi-GBM BGO detector response ($\lesssim 40\,\mathrm{MeV}$, where the effective area falls steeply) and the regime in which Fermi-LAT achieves robust detection efficiency ($\gtrsim 100\,\mathrm{MeV}$). The frequency $\nu_\star \simeq 37\,\mathrm{MeV}$ at which the foam-induced spectral break is predicted to be most constraining falls squarely within this gap, so neither instrument provides fully resolved coverage across the critical transition. Closing the MeV gap is a recognized community priority, addressed by next-generation satellite concepts that combine Compton-scattering and pair-conversion detection techniques to achieve continuous sensitivity from $\sim 100\,\mathrm{keV}$ to beyond 1 GeV. The All-sky Medium Energy Gamma-ray Observatory (AMEGO [25]) and its MIDEX-scale variant AMEGO-X [26] are designed to cover 100 keV to 1 GeV with order-of-magnitude improvements over both COMPTEL and Fermi-LAT in the MeV decade; the ESA-proposed M-class concept e-ASTROGAM [27] targets a comparable bandpass. The near-term NASA SMEX mission COSI (scheduled launch ∼2027) will extend broadband sensitivity to 0.2–5 MeV, providing a valuable pathfinder, though its upper energy bound falls well short of the 37 MeV regime and the gap therefore remains open. Realization of any of the broader-bandpass concepts would, for the first time, permit fully resolved spectral measurements of GRB prompt emission across the 37 MeV transition without the instrumental discontinuity that limits the present work; reducing the per-epoch line-width uncertainties to $\delta r \lesssim 0.01$ would tighten the bound to $\alpha \gtrsim 0.64$, removing a major systematic and enabling substantially tighter tests of spacetime-foam models. Independently, a joint fit of the line centroid and width across epochs — the present analysis constrains the width alone — would exploit the energy-dependent line-center shift $\nu_0^{\rm eff} = \nu_0\,\tau_G^2/(\tau_G^2 + \sigma_\Delta^2)$ (Eq. (17)) that the same foam jitter induces, adding a second handle on α and further tightening the bound.

A complementary implication runs in the opposite direction. The flagship goal of next-generation CMB spectrometers — PIXIE, SuperPIXIE, Voyage 2050 — is a first detection of the chemical potential μ, whose guaranteed ΛCDM value $\mu \simeq 2 \times 10^{-8}$ arises from Silk damping of primordial perturbations. Our Fisher analysis shows that the foam-induced distortion is nearly degenerate with the μ template across the accessible band (Section V B 3); the two cannot be separated from the monopole spectrum alone. A measured μ that *departs* from the ΛCDM prediction would be a landmark signal whatever its origin, and spacetime foam is one propagation-induced alternative to early-universe energy injection — distinguishable only with external information. The GRB lower bound $\alpha \gtrsim 0.63$ already limits any foam contribution to the CMB monopole to far below the ΛCDM μ, so foam does not contaminate the guaranteed-μ target; but this same spectroscopy-plus-GRB complementarity is what would be needed to attribute a larger, anomalous μ. In this sense the CMB μ-distortion measurement and the foam search are two readings of a single spectral degeneracy. The decisive limitation here is theoretical rather than instrumental: the pedestal shape — and hence how completely the foam deficit can be separated from a temperature shift or from μ — is fixed by the foam temporal autocorrelation $C_\alpha(\tau)$, which still awaits a first-principles derivation from a quantum-gravity model. The amplitude A itself is in the same position: it is calibrated phenomenologically rather than derived from a fluctuation-dissipation or holographic-entropy argument at the Planck-cell level.

The multi-epoch $r_{\rm obs}$ dataset constrains any propagation effect that broadens an emission line in proportion to a power of E, not only the Ng–Perlman spacetime-foam template considered here. The most natural extension is to stochastic Lorentz-invariance violation [14], in which the photon group velocity carries a Planck-suppressed random coefficient $v(E) = c\big[1 - \xi\,(E/E_{\rm QG,n})^n\big]$, with ξ of order unity drawn independently for each Planck-scale traversal. Photons in the same line then propagate at slightly different speeds, broadening the observed width as $\sigma_{\rm LIV}(E)/E \propto E^n$ on top of any intrinsic contribution. Unlike the deterministic LIV signal probed by photon time-of-flight studies [28] — where GRB 221009A LHAASO TeV photons already constrain $E_{\rm QG,1} \gtrsim$ a few $\times\, E_{\rm Pl}$ [19] — stochastic LIV produces a *width* signal that survives light-curve cross-correlation, and a multi-epoch $r_{\rm obs}$ measurement is therefore one of the few clean probes available. Fitting Eq. (28) recovers the foam constraint as the $n = 1$ case; refitting the same seven-epoch dataset under $r_{\rm LIV} \propto (E/E_0)^n$ would yield bounds on $\langle\xi^2\rangle^{1/2}/E^n_{\rm QG,n}$ at each order, with the $n = 2$ (quadratic) case particularly valuable because the deterministic time-of-flight tests are weakest there at $z = 0.151$.

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